

Task Manager Assessment Submission

PulseBoard is a polished Kanban-style task board designed for the internship assessment.

Candidate	Jacky Chen
Repository	[Add your GitHub repository URL here]
Live Demo	[Add your deployed frontend URL here]
Stack	React, TypeScript, Vite, Supabase, anonymous auth

1. Solution Overview

PulseBoard is a responsive four-column Kanban board that lets each guest user create tasks, move them across statuses, and visually manage work. The interface emphasizes strong hierarchy, clear spacing, fast scanability, and immediate feedback for loading, empty, and error states.

2. Design Decisions

- Used a glassmorphism-inspired dark surface with warm accent gradients to make the board feel intentional instead of generic.
- Kept drag-and-drop native with the HTML5 API to reduce dependency risk while preserving direct interaction.
- Added summary cards, due date urgency badges, and filtering so the board is more useful than a basic todo list.
- Included a demo fallback mode so the product can still be reviewed locally before Supabase credentials are added.

3. Features Implemented

- Anonymous guest session flow through Supabase Auth.
- Per-user task isolation with Row Level Security.
- Task creation with title, description, priority, and due date.
- Drag-and-drop status changes across To Do, In Progress, In Review, and Done.
- Search, priority filtering, and due date filtering.
- Summary statistics for total, completed, in-flight, overdue, and due-soon tasks.
- Realtime-ready subscription hook for task refresh when Supabase mode is enabled.

4. Database Schema

The full SQL setup is included in supabase/schema.sql. The main tasks table is summarized below.

Field	Type	Notes
id	uuid	Primary key
title	text	Required

status	text	todo, in_progress, in_review, done
description	text	Optional detail body, defaults to empty string
priority	text	low, normal, high
user_id	uuid	Tied to anonymous guest session via auth.uid()

Additional field included in the actual schema: created_at timestampz default timezone('utc', now()).

5. Setup Instructions

- Install dependencies with `pnpm install`.
- Copy `.env.example` to `.env` and set `VITE_SUPABASE_URL` plus `VITE_SUPABASE_ANON_KEY`.
- Enable Anonymous sign-in in Supabase Auth.
- Run the SQL in `supabase/schema.sql`.
- Start locally with `pnpm dev`.
- Deploy the Vite app to Vercel, Netlify, or Cloudflare Pages with the same two environment variables.

6. Tradeoffs and Next Steps

- I prioritized a stable, reviewer-friendly core board over adding a larger optional backend surface.
- If I had more time, I would add assignees, task comments, and a dedicated activity log.
- I would also add integration tests around anonymous auth bootstrapping and drag-and-drop persistence.